

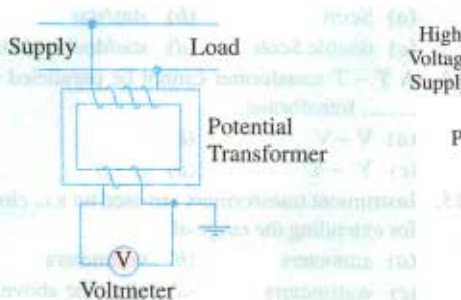


Fig. 33.34

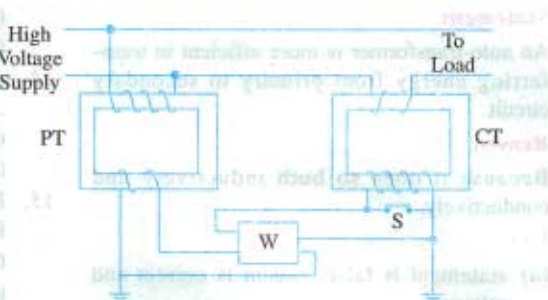


Fig. 33.35

Fig. 33.35 shows the connections of instrument transformers to a wattmeter. While connecting the wattmeter, the relative polarities of the secondary terminals of the transformers with respect to their primary terminals must be known for connections of the instruments.

OBJECTIVE TEST – 33

- Which of the following connections is best suited for 3-phase, 4-wire service ?
(a) $\Delta - \Delta$ (b) Y-Y
(c) $\Delta - Y$ (d) Y- Δ
- In a three-phase Y-Y transformer connection, neutral is fundamental to the
(a) suppression of harmonics
(b) passage of unbalanced currents due to unbalanced loads
(c) provision of dual electric service
(d) balancing of phase voltages with respect to line voltages.
- As compared to $\Delta - \Delta$ bank, the capacity of the V-V bank of transformers is percent.
(a) 57.7 (b) 66.7
(c) 50 (d) 86.6
- If three transformers in a $\Delta - \Delta$ are delivering their rated load and one transformer is removed, then overload on *each* of the remaining transformers is percent.
(a) 66.7 (b) 173.2
(c) 73.2 (d) 58
- When a V-V system is converted into a $\Delta - \Delta$ system, increase in capacity of the system is percent.
(a) 86.6 (b) 66.7
(c) 73.2 (d) 50
- For supplying a balanced 3- ϕ load of 40-kVA, rating of each transformer in V-V bank should be nearly kVA.
(a) 20 (b) 23
(c) 34.6 (d) 25
- When a closed- Δ bank is converted into an open- Δ bank, each of the two remaining transformers supplies percent of the original load.
(a) 66.7 (b) 57.7
(c) 50 (d) 73.2
- If the load p.f. is 0.866, then the average p.f. of the V-V bank is
(a) 0.886 (b) 0.75
(c) 0.51 (d) 0.65
- A T-T connection has higher ratio of utilization than a V-V connection only when
(a) identical transformers are used
(b) load power factor is leading
(c) load power factor is unity
(d) non-identical transformers are used.
- The biggest advantage of T-T connection over the V-V connection for 3-phase power transformation is that it provides
(a) a set of balanced voltages under load
(b) a true 3-phase, 4-wire system
(c) a higher ratio of utilization
(d) more voltages.
- Of the following statements concerning parallel operation of transformers, the one which is **not** correct is
(a) transformers must have equal voltage ratings
(b) transformers must have same ratio of transformation
(c) transformers must be operated at the same frequency
(d) transformers must have equal kVA ratings.

12. Statement

An auto-transformer is more efficient in transferring energy from primary to secondary circuit.

Reason

Because it does so both inductively and conductively.

Key

- (a) statement is false, reason is correct and relevant
 (b) statement is correct, reason is correct but irrelevant
 (c) both statement and reason are correct and are connected to each other as cause and effect
 (d) both statement and reason are false.
- 13.** Out of the following given choices for poly phase transformer connections which one will you select for three-to-two phase conversion ?

- (a) Scott (b) star/star
 (c) double Scott (d) star/double-delta

14. A T – T transformer cannot be paralleled with transformer.

- (a) V – V (b) Y – Δ
 (c) Y – Y (d) Δ – Δ

15. Instrument transformers are used on a.c. circuits for extending the range of

- (a) ammeters (b) voltmeters
 (c) wattmeters (d) all of the above.

16. Before removing the ammeter from a current transformer, its secondary must be short-circuited in order to avoid

- (a) excessive heating of the core
 (b) high secondary e.m.f.
 (c) increase in iron losses
 (d) all of the above.

ANSWERS

- 1. c 2. a 3. a 4. c 5. c 6. b 7. b 8. b 9. d 10. b 11. d 12. c 13. a 14. b 15. d 16. d**